

ECON 6290: Labor Markets in Developing Countries

Week 5
Fall 2026

Lisa Ho

Announcements

1. Mini-proposal #1

- Due next Friday (**October 16**)

2. Shape of the day: broadly 'reconciling domestic and market work'

- First half: time use, children
- Break (5 minutes)
- Discussion: West Bengal paper (I'm leading this week)
- Experiment: impacts of childcare provision (Egypt)

3. Any questions?

Most women around the world say they would like to be in paid work

- **ILO–Gallup World Poll (2017):** $\approx 149,000$ adults aged 15+ in 142 countries.
 - “Would you *prefer* to work at a paid job, or stay at home and take care of your family and the housework, or would you prefer to do both?”

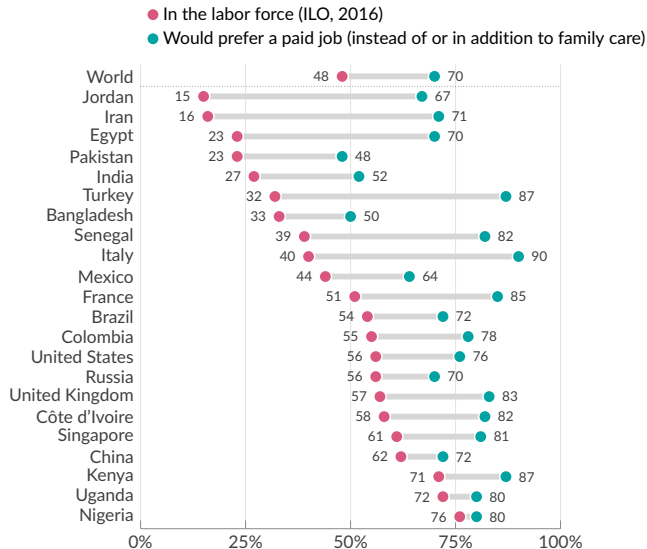
Most women around the world say they would like to be in paid work

- **ILO-Gallup World Poll (2017):** $\approx$ 149,000 adults aged 15+ in 142 countries.
 - “Would you *prefer* to work at a paid job, or stay at home and take care of your family and the housework, or would you prefer to do both?”
- Worldwide, **70%** of women would prefer a paid job (29%) or **both** a paid job and caring for their family (41%); 27% would prefer to stay at home
 - Share of women who prefer paid work much exceeds share of women in labor force
 - Motivates what I call “latent labor force” of women who prefer paid work

Most women around the world say they would like to be in paid work

- **ILO–Gallup World Poll (2017):** $\approx$ 149,000 adults aged 15+ in 142 countries.
 - “Would you *prefer* to work at a paid job, or stay at home and take care of your family and the housework, or would you prefer to do both?”
- Worldwide, **70%** of women would prefer a paid job (29%) or **both** a paid job and caring for their family (41%); 27% would prefer to stay at home
 - Share of women who prefer paid work much exceeds share of women in labor force
 - Motivates what I call “latent labor force” of women who prefer paid work
- Gallup meant to ask men the same question about themselves
 - Say they dropped it because in piloting, outside a handful of countries, “the question did not resonate with men. Many men were incredulous when asked the question.”
 - Instead asked men what they would prefer the women in their households do

...This “latent labor force” is especially large in MENA and South Asia



If they want paid work... why don't they try to get it?

- One (according to me) of the most important reasons is (as in the paper for today):
 1. Gender norms create a division of labor within the home where women are expected to do most of the unpaid work at home – cooking, cleaning, care for children and the elderly
 2. Market work has to be **reconciled** with that domestic labor; often that's $\approx$ impossible

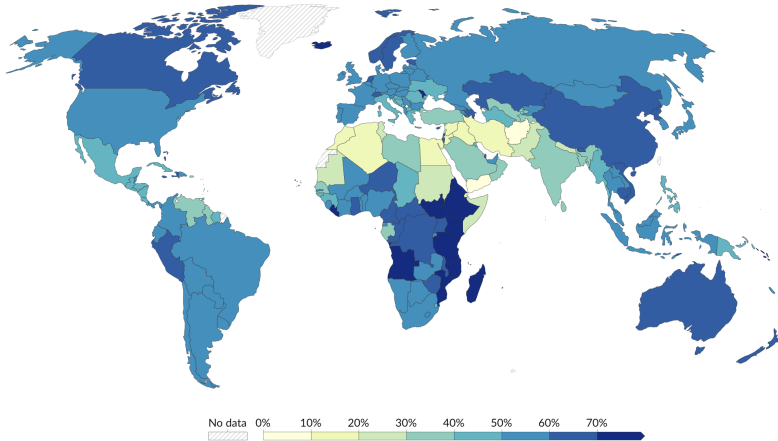
If they want paid work... why don't they try to get it?

- One (according to me) of the most important reasons is (as in the paper for today):
 1. Gender norms create a division of labor within the home where women are expected to do most of the unpaid work at home – cooking, cleaning, care for children and the elderly
 2. Market work has to be **reconciled** with that domestic labor; often that's $\approx$ impossible
- **Today:** the challenge of reconciling home production with market work
 1. First, some basic facts about FLFP across countries and over time
 2. Challenges of location: if women are expected to stay at home, job has to come to them (Andrew and Smurra)
 3. Challenges of children: work on the child penalty (Kleven, Landais and Leite-Mariante)
 4. Challenges of gendered norms of home production: housework does not shift to men even when she earns more (Hancock, Lafortune and Low)

Reconciling domestic and market work

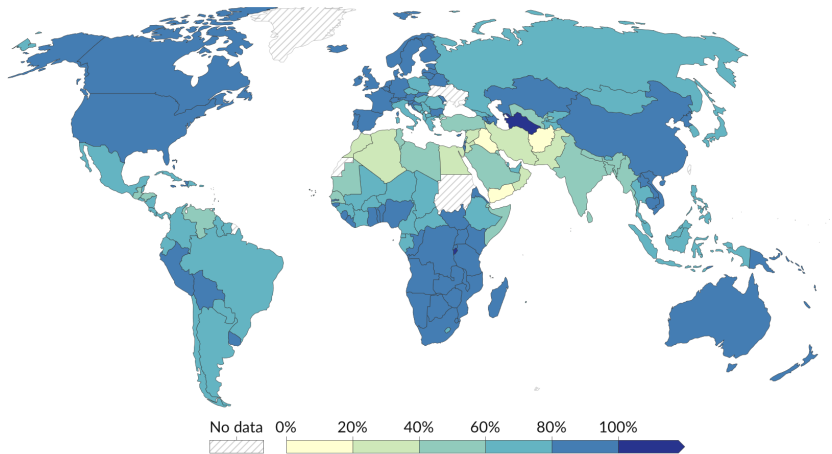
1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

Large disparities in female labor force participation across the world (ILO, 2025)



Female labor force participation (FLFP): % of working-age women employed or seeking work

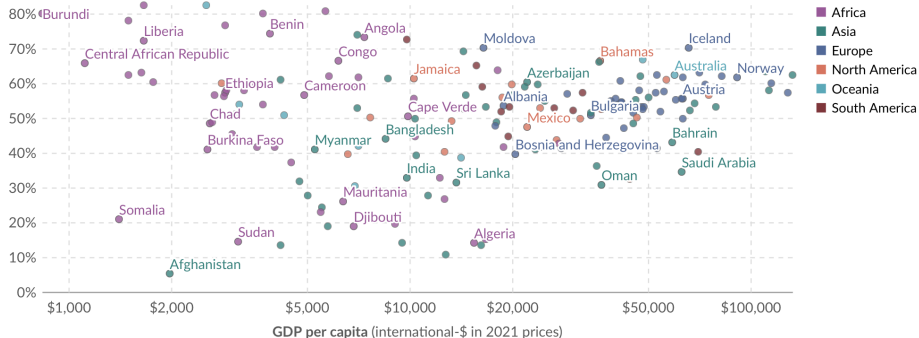
Also in the **ratio of female to male labor force participation** (ILO, 2025)



Nearly everywhere, **male LFP** > **female LFP** (and the ratio, like the level, is lowest in South Asia, the Middle East and North Africa)

Across countries, FLFP is U-shaped in GDP per capita (ILO, 2024)

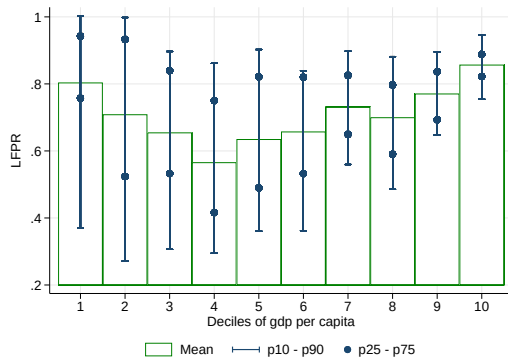
Female labor force participation rate (%) (% of female population ages 15+, modeled ILO estimate)



What generate the U? Goldin (1995), "The U-shaped female labor force function in economic development and economic history".

- **Low income:** women work out of necessity, mostly on family farms and in household enterprises
- **Middle income:** household income rises, and work moves off the farm and out of the home → income effect dominates, women exit
- **High income:** women's wages and education rise → substitution effect dominates, women re-enter

The female/male ratio follows the same U, with wide spread



- Mean female/male LFP ratio by decile of GDP per capita: the same U as the level
- But within every decile there are country-cohorts with a ratio between 0.85 and 1 (the 90th percentile)

Source: Ashraf, Bandiera, Minni and Quintas-Martínez (2026)

But the U is mostly a cross-country pattern, not a path countries follow

- The U hypothesis is about how FLFP changes *within* a country as it develops, but the evidence for it is mostly cross-sectional

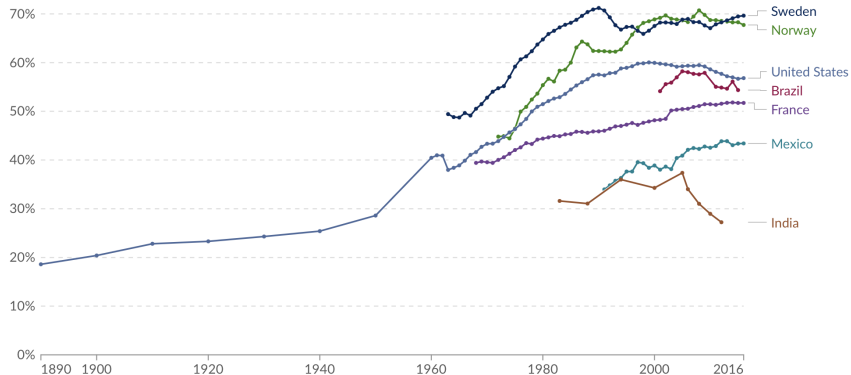
But the U is mostly a cross-country pattern, not a path countries follow

- The U hypothesis is about how FLFP changes *within* a country as it develops, but the evidence for it is mostly cross-sectional
- **Gaddis and Klasen (2014)**: with country fixed effects (identified only from changes within countries over time), support for the U is “feeble and depends on the data sources used”, and it vanishes in dynamic panels
- **Klasen (2019)**: most of the variation in FLFP across countries is in the country fixed effects, not in a country's position along the U
- **Olivetti (2014)**: today's rich countries did trace a U from 1890 on; countries developing after 1950 mostly have not

But the U is mostly a cross-country pattern, not a path countries follow

- The U hypothesis is about how FLFP changes *within* a country as it develops, but the evidence for it is mostly cross-sectional
- **Gaddis and Klasen (2014)**: with country fixed effects (identified only from changes within countries over time), support for the U is “feeble and depends on the data sources used”, and it vanishes in dynamic panels
- **Klasen (2019)**: most of the variation in FLFP across countries is in the country fixed effects, not in a country's position along the U
- **Olivetti (2014)**: today's rich countries did trace a U from 1890 on; countries developing after 1950 mostly have not

FLFP over time: increasing in most places, then stagnating

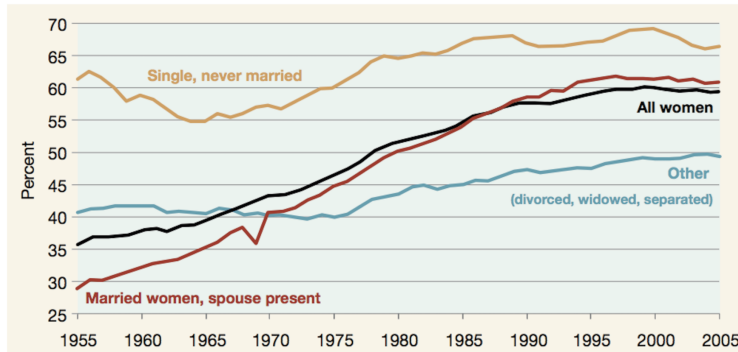


- U.S. FLFP rose from under 20% in 1890 to 60% by 2000
- Sweden and Norway rose fastest after 1960
- Many countries have stagnated since the 1990s (U.S., Brazil), and India *fell* after 2005 (but is again on the rise)

Source: Our World in Data. FLFP, % of women aged 15+.

In the U.S., the increase was driven by *married* women

Labor Force Participation of Women in the USA, 1955-2005



Married women's LFP doubled from 29% in 1955 to >60% by 2000; single women's barely changed
Hints at what might have been the binding constraint that was removed

Reconciling domestic and market work

1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

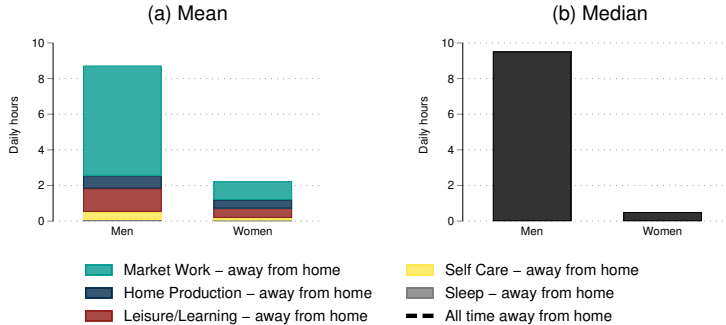
How much time do women spend outside their home?

- Gendered home production means women have to spend more time at home
 - On top of that, in some places female seclusion as a way to maintain purity and respect
- **Andrew and Smurra (2026)** measure **realized seclusion**: hours per day a person spends outside their own dwelling in India
 - Doesn't try to answer why, e.g. permission to leave, norms, or safety: it captures the net effect of all channels
 - Can compare across groups and over time
- **Data**: India Time Use Survey
 - **2019**: nationally representative, 138,798 households and 445,299 individuals
 - **1998**: pilot in six states, 63,269 individuals
 - 24-hour recall diary in 30-minute slots; for each activity, whether it was done inside or outside the dwelling

How much time do women spend outside their home?

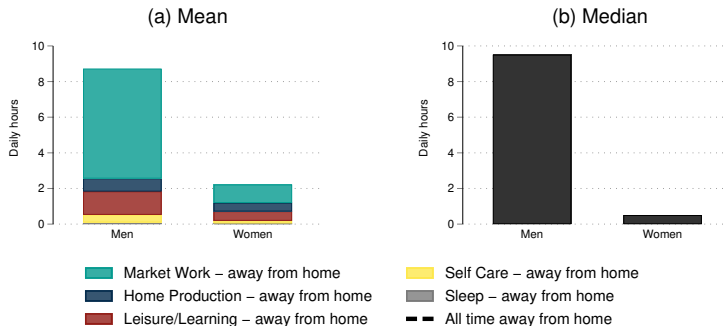
- Gendered home production means women have to spend more time at home
 - On top of that, in some places female seclusion as a way to maintain purity and respect
- **Andrew and Smurra (2026)** measure **realized seclusion**: hours per day a person spends outside their own dwelling in India
 - Doesn't try to answer why, e.g. permission to leave, norms, or safety: it captures the net effect of all channels
 - Can compare across groups and over time
- **Data**: India Time Use Survey
 - **2019**: nationally representative, 138,798 households and 445,299 individuals
 - **1998**: pilot in six states, 63,269 individuals
 - 24-hour recall diary in 30-minute slots; for each activity, whether it was done inside or outside the dwelling
- Main sample: currently married women and men aged 18–64 ($N = 245,743$)

Fact 1: Married women spend a median of 30 minutes a day outside their home



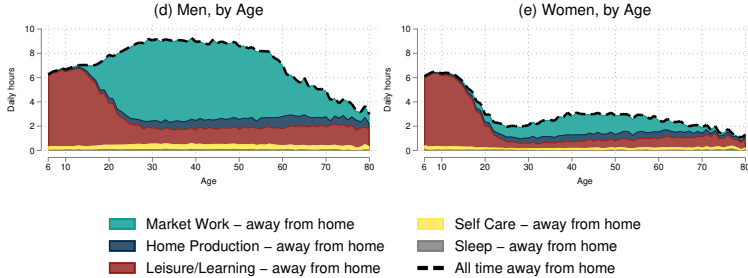
- Married adults 18–64, 2019. Women: mean **2.2 hours**, median **0.5 hours**; **45%** do not leave home on the diary day. Men: mean 8.7, median 9.5 hours; 4% do not leave home

Fact 1: Married women spend a median of 30 minutes a day outside their home

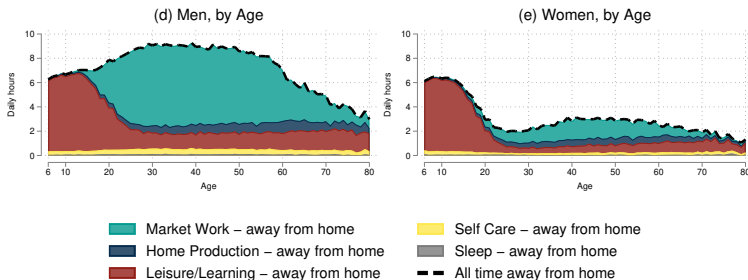


- Married adults 18–64, 2019. Women: mean **2.2 hours**, median **0.5 hours**; **45%** do not leave home on the diary day. Men: mean 8.7, median 9.5 hours; 4% do not leave home
- **Not only because women do less market work:** among those who did no market work that day, women spend 1.2 hours outside vs. 5.1 for men; 55% of these women vs. 15% of these men do not leave home

Fact 2:



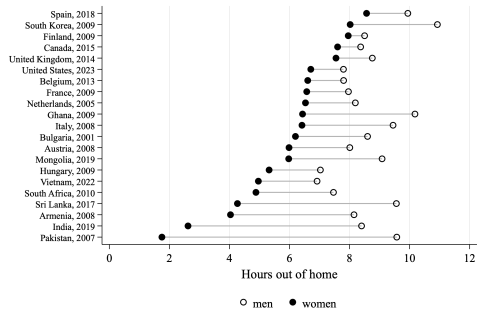
Fact 2:



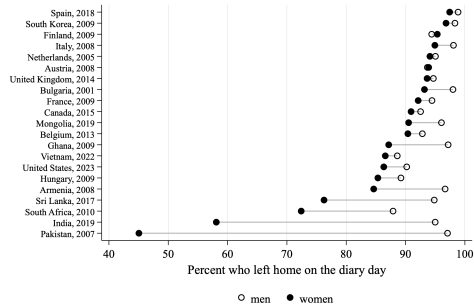
- No gap in childhood (about 6 hours a day outside for both); girls' time outside starts falling around **age 14**; the gap peaks at about **7 hours at age 30**
- Women aged 25 are the most secluded: 2 hours outside, 51% don't leave home on a given day
- At every age, married women are more secluded than unmarried women
- **Childcare explains little**: the gap is 6.1 hours in households with no children; a child aged 0–5 adds 0.9 hours

India and Pakistan are outliers in women's time outside the home

(a) Hours outside the home



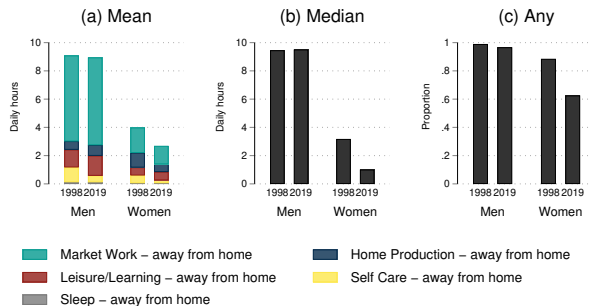
(b) Percent who left home on the diary day



- Adults 18–65 in 20 countries: men spend 6.9–10.9 hours a day outside everywhere
- Women: India 2.6 hours, Pakistan 1.8 hours vs. 4.0–8.6 hours in other countries; 42% (India) and 55% (Pakistan) do not leave home that day vs. at most 28% elsewhere

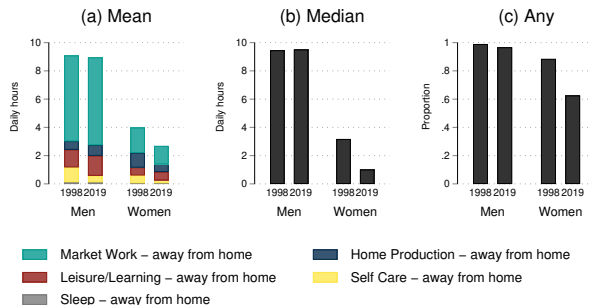
Time-use surveys (MTUS and national microdata).

Fact 3: Seclusion rose between 1998 and 2019



- Married women, six states: mean time outside fell from 3.98 to **2.68 hours**; share not leaving home rose from 12% to **38%**. Men unchanged
- Richer, more educated women are the most secluded, but poorer women have caught up with rich women's 1998 level

Fact 3: Seclusion rose between 1998 and 2019



- Married women, six states: mean time outside fell from 3.98 to **2.68 hours**; share not leaving home rose from 12% to **38%**. Men unchanged
- Richer, more educated women are the most secluded, but poorer women have caught up with rich women's 1998 level
- Is rising seclusion necessarily bad?
 - Certainly not all of it: toilets, piped water and cooking gas moved chores indoors
 - Composition explains 44% of the fall (type of dwelling alone 18%)

Reconciling domestic and market work

1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

The Child Penalty Atlas

- **Question:** How large are child penalties in employment around the world, and how do they change with economic development?
- **Child penalty:** the percentage by which women fall behind men in employment after the birth of their first child

The Child Penalty Atlas

- **Question:** How large are child penalties in employment around the world, and how do they change with economic development?
- **Child penalty:** the percentage by which women fall behind men in employment after the birth of their first child
- **Data:** 134 countries, 95.5% of the world's population, about 281 million person-year observations
 - Only 7 countries have panel data (Nordic registers, Austria, Switzerland, Australia, Japan)
 - The other 127 have only cross-sections: 77 censuses (mostly IPUMS), 17 DHS, 16 LIS, 8 EU-SILC, 5 labor force surveys (all low-income countries), 4 other

The Child Penalty Atlas

- **Question:** How large are child penalties in employment around the world, and how do they change with economic development?
- **Child penalty:** the percentage by which women fall behind men in employment after the birth of their first child
- **Data:** 134 countries, 95.5% of the world's population, about 281 million person-year observations
 - Only 7 countries have panel data (Nordic registers, Austria, Switzerland, Australia, Japan)
 - The other 127 have only cross-sections: 77 censuses (mostly IPUMS), 17 DHS, 16 LIS, 8 EU-SILC, 5 labor force surveys (all low-income countries), 4 other
- **Outcome:** employment only (usually “worked last week”), no earnings or hours
- **Methodological contribution:** an event study you can run with a single cross-section

The child penalty event study

- With panel data, estimate separately for women and men ($g \in \{w, m\}$):

$$Y_{it}^g = \alpha^g \cdot \mathbf{D}_{it}^{\text{Event}} + \beta^g \cdot \mathbf{D}_{it}^{\text{Age}} + \gamma^g \cdot \mathbf{D}_{it}^{\text{Year}} + \nu_{it}^g$$

- Event time t is relative to the first birth; $t = -2$ (the year before pregnancy) omitted; age and year dummies absorb life-cycle and time trends

The child penalty event study

- With panel data, estimate separately for women and men ($g \in \{w, m\}$):

$$Y_{it}^g = \alpha^g \cdot \mathbf{D}_{it}^{\text{Event}} + \beta^g \cdot \mathbf{D}_{it}^{\text{Age}} + \gamma^g \cdot \mathbf{D}_{it}^{\text{Year}} + \nu_{it}^g$$

- Event time t is relative to the first birth; $t = -2$ (the year before pregnancy) omitted; age and year dummies absorb life-cycle and time trends
- Scale each event-time coefficient by counterfactual employment, $P_t^g = \hat{\alpha}_t^g / \mathbb{E}[\tilde{Y}_{it}^g \mid t]$, where $\tilde{Y}$ is the prediction without the event dummies
- **Child penalty** = average gap in the scaled effects over the 10 years after birth, net of the pre-period:

$$\text{Child Penalty} = \mathbb{E}[P_t^m - P_t^w \mid t \geq 0] - \mathbb{E}[P_t^m - P_t^w \mid t < 0]$$

The child penalty event study

- With panel data, estimate separately for women and men ($g \in \{w, m\}$):

$$Y_{it}^g = \alpha^g \cdot \mathbf{D}_{it}^{\text{Event}} + \beta^g \cdot \mathbf{D}_{it}^{\text{Age}} + \gamma^g \cdot \mathbf{D}_{it}^{\text{Year}} + \nu_{it}^g$$

- Event time t is relative to the first birth; $t = -2$ (the year before pregnancy) omitted; age and year dummies absorb life-cycle and time trends
- Scale each event-time coefficient by counterfactual employment, $P_t^g = \hat{\alpha}_t^g / \mathbb{E}[\tilde{Y}_{it}^g \mid t]$, where $\tilde{Y}$ is the prediction without the event dummies
- **Child penalty** = average gap in the scaled effects over the 10 years after birth, net of the pre-period:

$$\text{Child Penalty} = \mathbb{E}[P_t^m - P_t^w \mid t \geq 0] - \mathbb{E}[P_t^m - P_t^w \mid t < 0]$$

- **Identifying assumption:** the exact timing of the first birth is orthogonal to other dynamics in women's employment relative to men's

Without panel data: the pseudo-event study (Kleven 2023)

- **Problem:** in a cross-section, the age of the oldest co-resident child places parents at $t \geq 0$. But we can't place childless people at $t < 0$: we don't know if or when they will have a first child

Without panel data: the pseudo-event study (Kleven 2023)

- **Problem:** in a cross-section, the age of the oldest co-resident child places parents at $t \geq 0$. But we can't place childless people at $t < 0$: we don't know if or when they will have a first child
- **Solution:** give each parent stand-in “pre-birth” observations from childless people
 - Parent at $t = 0$, age a , year $y \rightarrow$ the stand-in for $t = -n$ is childless people aged $a - n$ in year $y - n$ (same birth cohort), with the same sex, education, marital status and urban/rural status
 - Repeat for $n = 1, \dots, 5$ to get a pseudo-panel, then run the same event study
 - With only one survey year, the stand-ins are younger childless people in the *same* year y

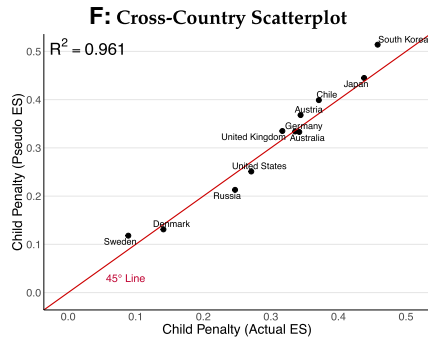
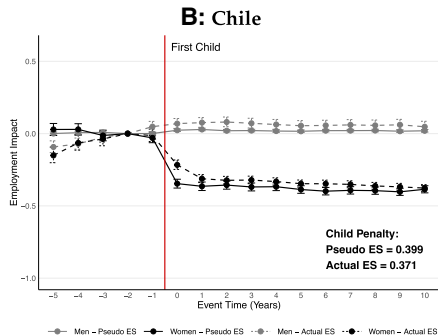
Without panel data: the pseudo-event study (Kleven 2023)

- **Problem:** in a cross-section, the age of the oldest co-resident child places parents at $t \geq 0$. But we can't place childless people at $t < 0$: we don't know if or when they will have a first child
- **Solution:** give each parent stand-in “pre-birth” observations from childless people
 - Parent at $t = 0$, age a , year $y \rightarrow$ the stand-in for $t = -n$ is childless people aged $a - n$ in year $y - n$ (same birth cohort), with the same sex, education, marital status and urban/rural status
 - Repeat for $n = 1, \dots, 5$ to get a pseudo-panel, then run the same event study
 - With only one survey year, the stand-ins are younger childless people in the *same* year y
- **Q:** What has to be true for this to work?

Without panel data: the pseudo-event study (Kleven 2023)

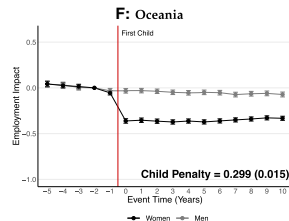
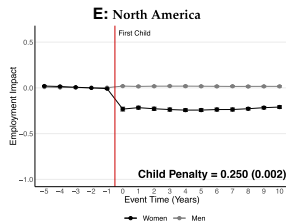
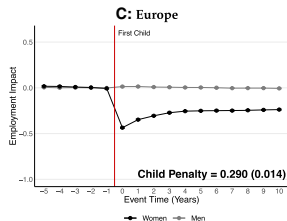
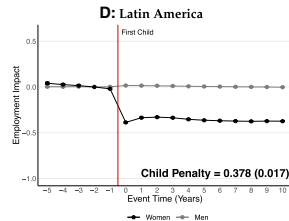
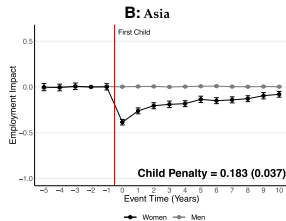
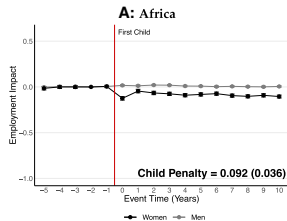
- **Problem:** in a cross-section, the age of the oldest co-resident child places parents at $t \geq 0$. But we can't place childless people at $t < 0$: we don't know if or when they will have a first child
- **Solution:** give each parent stand-in “pre-birth” observations from childless people
 - Parent at $t = 0$, age a , year $y \rightarrow$ the stand-in for $t = -n$ is childless people aged $a - n$ in year $y - n$ (same birth cohort), with the same sex, education, marital status and urban/rural status
 - Repeat for $n = 1, \dots, 5$ to get a pseudo-panel, then run the same event study
 - With only one survey year, the stand-ins are younger childless people in the *same* year y
- **Q:** What has to be true for this to work?
 - Given the matching variables, childless people aged $a - n$ have the same employment as people who will have their first child at a . Fails if, e.g., women who never have children are more career-focused: pre-period too high, penalty overstated
 - With one survey year: today's 25-year-olds look like today's 30-year-olds did at 25 (no cohort-specific shocks)

Validation: the pseudo and true event studies line up



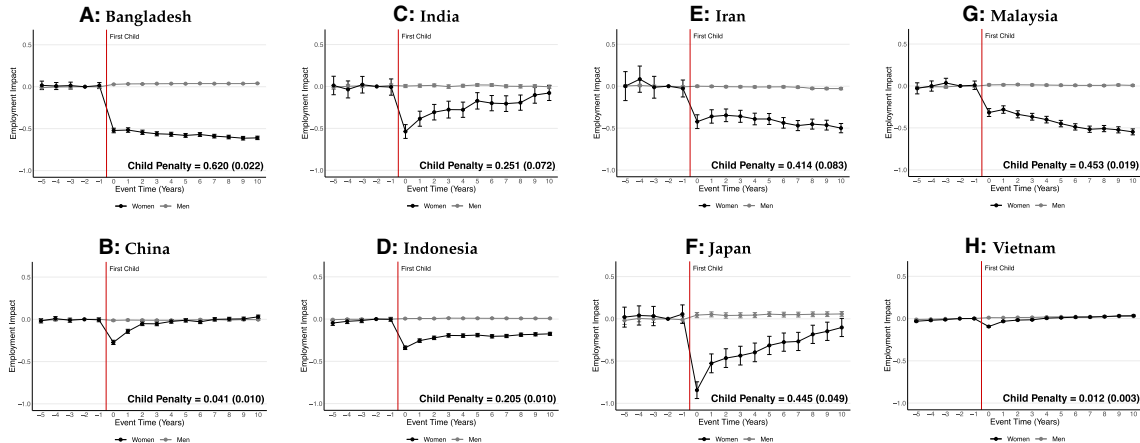
- In 12 countries with panel data, re-estimate using only the cross-sectional information:
 $R^2 = 0.96$ against the true event studies
- But the 12 are high- and middle-income countries (Chile is the poorest). No validation where most of the atlas comes from

Child penalties by continent: smallest in Africa, largest in Latin America



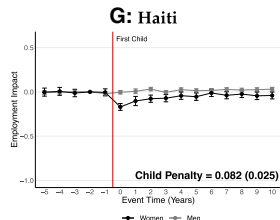
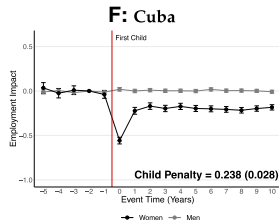
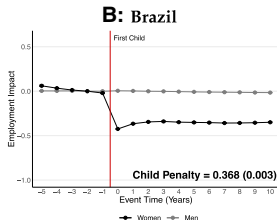
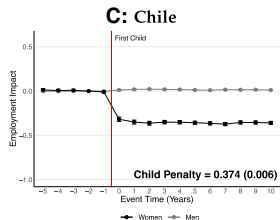
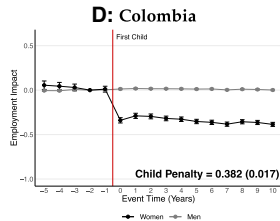
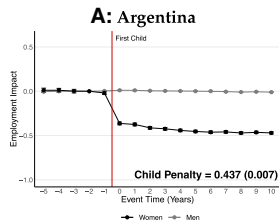
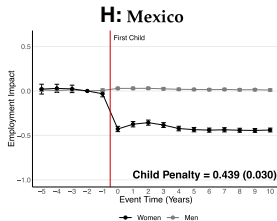
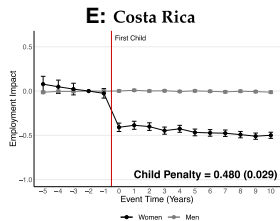
Africa 9%, Asia 18%, North America 25%, Europe 29%, Oceania 30%, Latin America 38%

Asia: from zero to 62% (India: 25%)



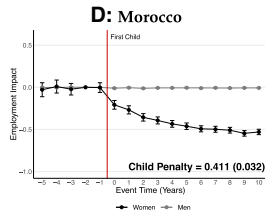
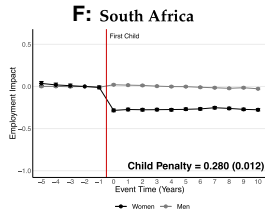
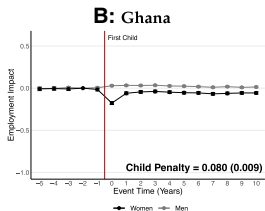
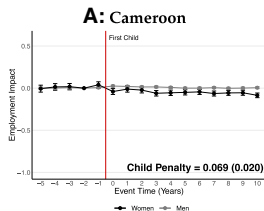
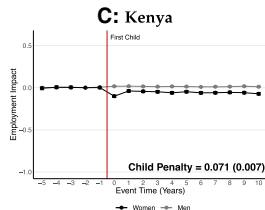
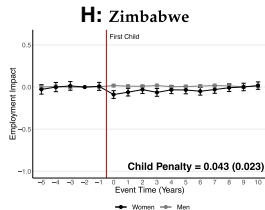
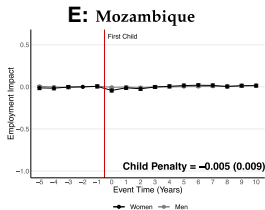
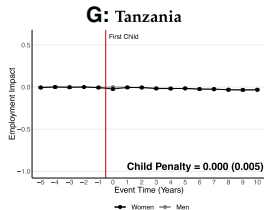
Bangladesh 62%, Malaysia 45%, Japan 45%, Iran 41%, India 25% (s.e. 7 pp), Indonesia 21%, China 4%, Vietnam 1%

Latin America: consistently large, except in the Caribbean



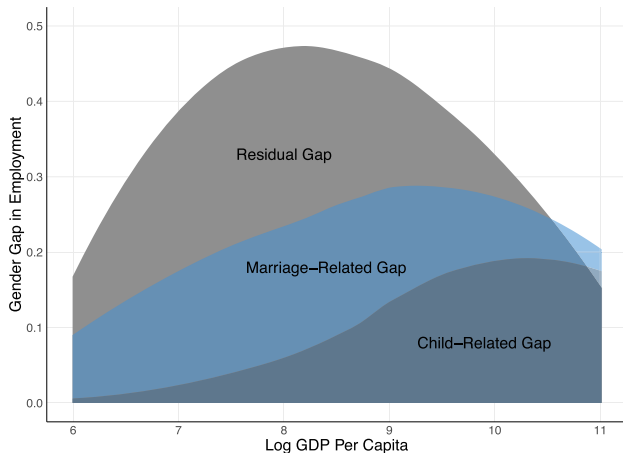
Costa Rica 48%, Mexico 44%, Argentina 44%, Colombia 38%, Chile 37%, Brazil 37% vs. Cuba 24%, Haiti 8%

Africa: near zero in the poorest countries, large in Morocco



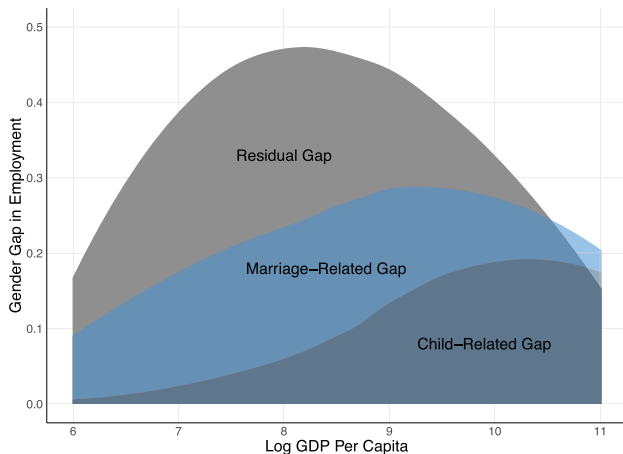
Tanzania 0%, Mozambique -0.5%, Zimbabwe 4%, Kenya 7%, Ghana 8% vs. South Africa 28%, Morocco 41%

With development, gender gaps shift from marriage to children



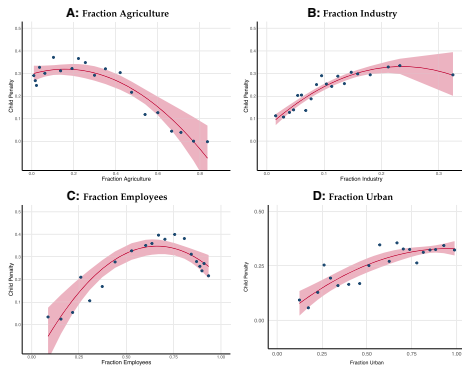
- Total employment gap: an **inverted U** in GDP per capita (the mirror of the FLFP U)
- Part due to **children**: near zero in poor countries, rises with income
- Part due to **marriage**: large in poor and middle-income countries, fades at high income

With development, gender gaps shift from marriage to children



- Total employment gap: an **inverted U** in GDP per capita (the mirror of the FLFP U)
- Part due to **children**: near zero in poor countries, rises with income
- Part due to **marriage**: large in poor and middle-income countries, fades at high income
- Family formation explains about **half** the gap in low- and middle-income countries, **all** of it in rich ones
- Caveat: this split uses a static regression, not the event study

Mechanism: structural transformation separates the home from the workplace

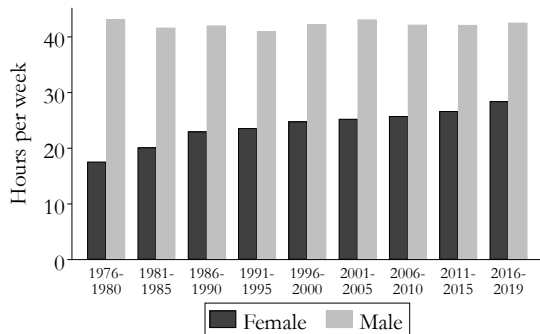


- Penalties fall with the agricultural share, and rise with industry, salaried work and urbanization; cities have larger penalties than their countries (Nairobi 22% vs. Kenya 7%)
- U.S. history: the (unscaled) penalty was 5 pp in 1880–1910 and 43 pp in 1950–70 — “the stay-at-home mom is a relatively modern phenomenon”

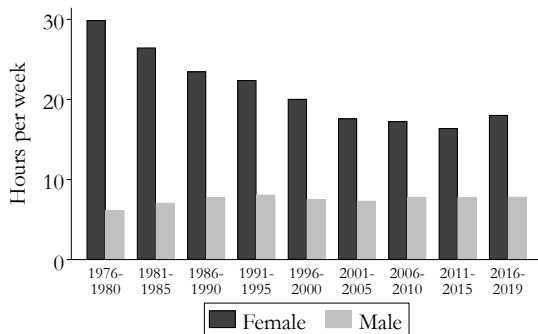
Reconciling domestic and market work

1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

Women's market hours rose and their housework fell; men's housework has not moved



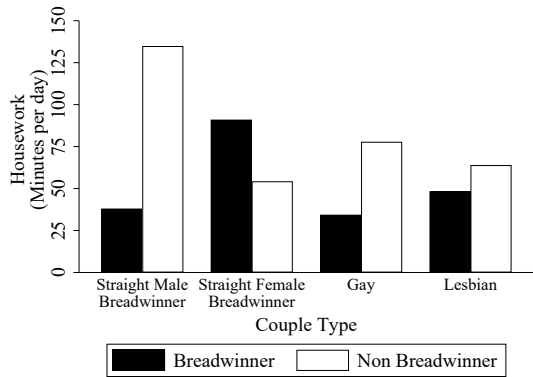
(a) Work Hours



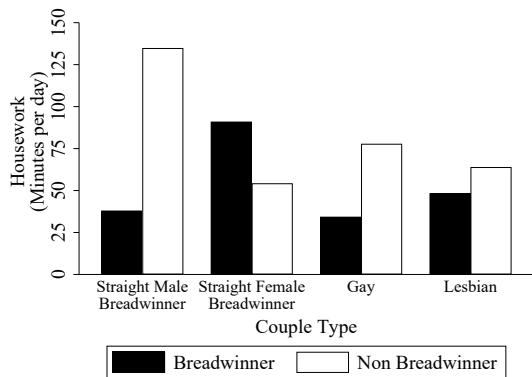
(b) Housework Hours

- PSID, 1976–2019: women's weekly market hours rose and their housework fell; men's housework "has remained completely flat since the 1980s"

PSID, hours per week.

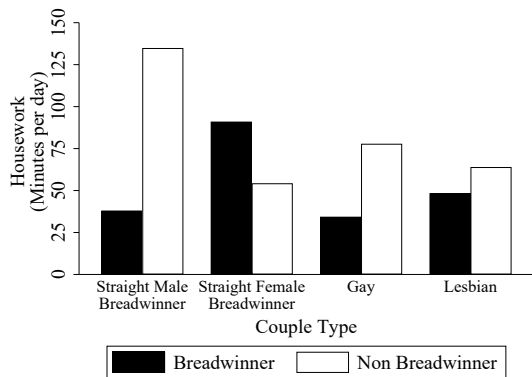


Female breadwinners still do more housework



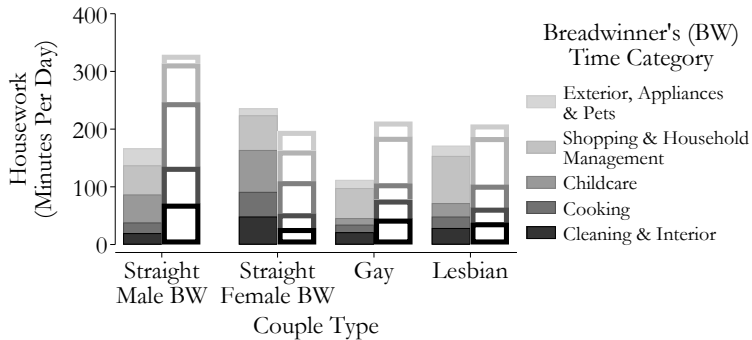
- U.S. time use (ATUS 2003–2019)
- The higher earner does *less* housework in male-breadwinner, gay and lesbian couples

Female breadwinners still do more housework



- U.S. time use (ATUS 2003–2019)
- The higher earner does *less* housework in male-breadwinner, gay and lesbian couples
- The exception: straight couples where the woman earns more. The lower-earning man does **44 fewer minutes a day** of housework than she does (25 with couple fixed effects in the PSID)
- Cooking and cleaning, not childcare; holds for childless couples

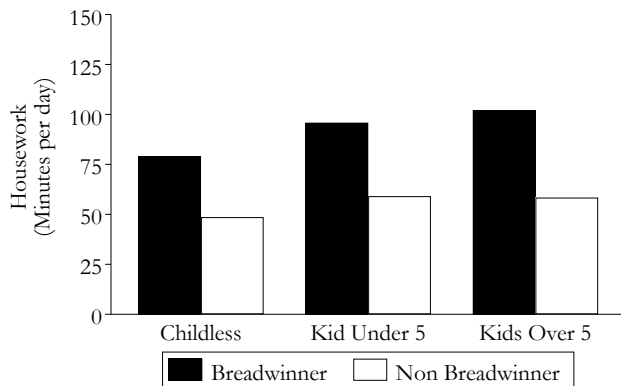
The gap is in cooking and cleaning, not childcare



- Cooking and cleaning are a large share of home production, and “are performed infrequently by all breadwinners except heterosexual female breadwinners”
- Men do more exterior and appliance maintenance, but it is a small share of the total

ATUS 2003–2019; solid bars = breadwinner, hollow = secondary earner.

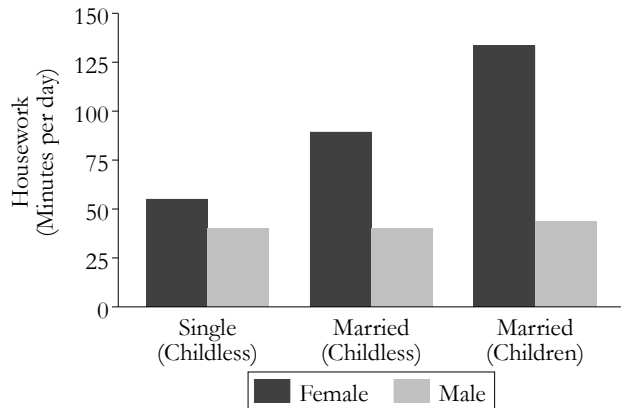
The reversal holds with and without children



ATUS 2003–2019, minutes per day.

- Straight couples where the woman earns more: she does more housework than he does whether they are childless, have a child under 5, or children over 5

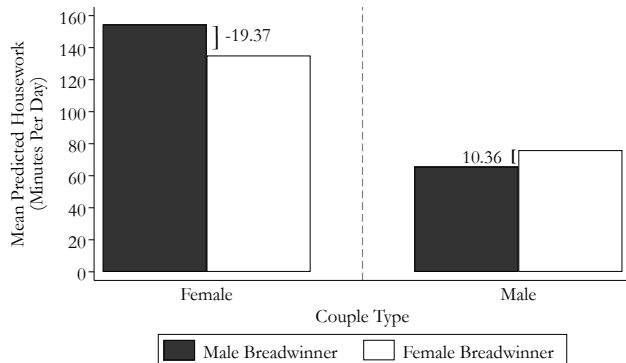
Women's housework jumps at marriage, even without children



ATUS 2003–2019, minutes per day.

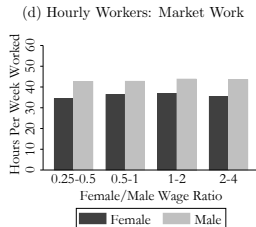
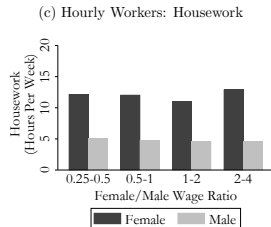
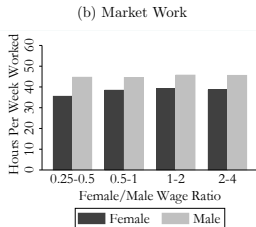
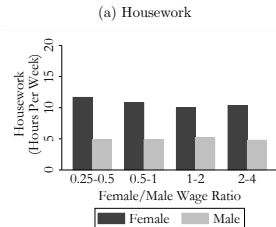
- Single, childless women do only marginally more housework than single men
- Women's housework "balloons after entering into marriage, *even if no children are present*"; men's is flat across the three groups
- Points to specialization within the household rather than different preferences

Within couples: when she becomes the breadwinner, little changes



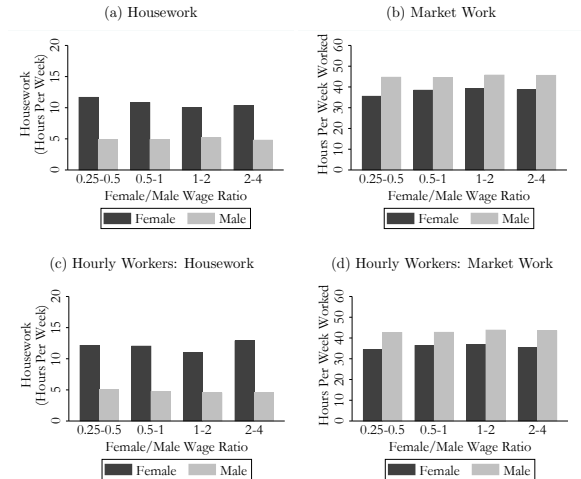
- PSID couples who switch from a male to a female breadwinner, with couple fixed effects
- Her housework falls by 19 minutes a day; his rises by 10 – she still does much more

Dual earners: men's hours barely move with relative wages



- Women's housework falls as their relative wage rises, but stays above men's in every group; men's market hours stay just above 40
- Why look at hourly workers?

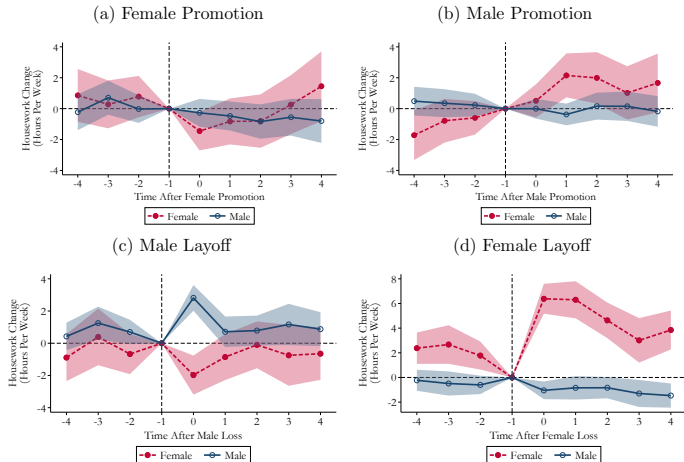
Dual earners: men's hours barely move with relative wages



- Women's housework falls as their relative wage rises, but stays above men's in every group; men's market hours stay just above 40
- Why look at hourly workers? Same pattern when both partners are paid by the hour (bottom panels), so men could adjust their hours

ATUS 2003-2019; female/male wage ratio groups 0.25-0.5, 0.5-1, 1-2, 2-4.

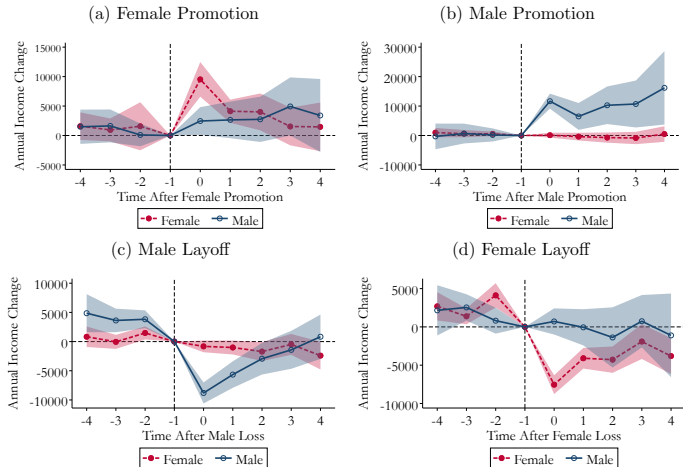
Promotions and layoffs



- When she is promoted, her housework falls slightly and his does not change; when he is promoted, her housework rises
- When he is laid off, his housework rises briefly and returns to baseline; when she is laid off, hers rises by much more and stays up

PSID event studies, hours per week; 382–1,052 couples per panel.

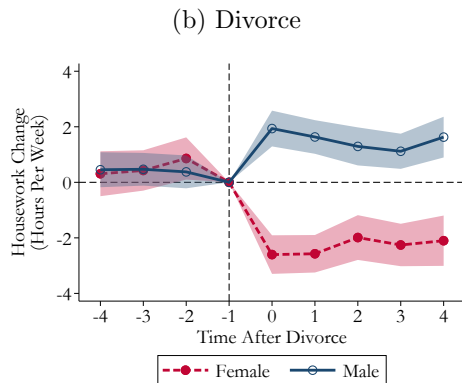
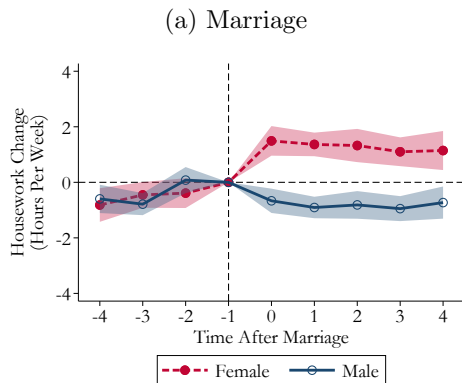
These are real earnings shocks



PSID, change in annual income.

- Promotions raise annual earnings by about \$5,000 (women) and \$10,000 (men) for at least three years; layoffs cause similar drops
- Laid-off men's earnings take four years to recover, but their housework is back to baseline much sooner

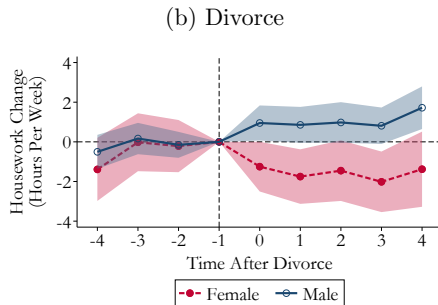
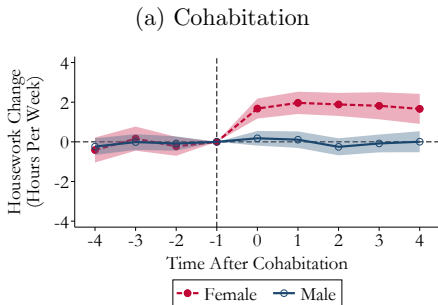
Effects of marriage and divorce



- PSID event studies: at marriage, women's housework rises and men's falls; after divorce, women's falls and men's rises

Change in housework, hours per week, relative to the year before the event.

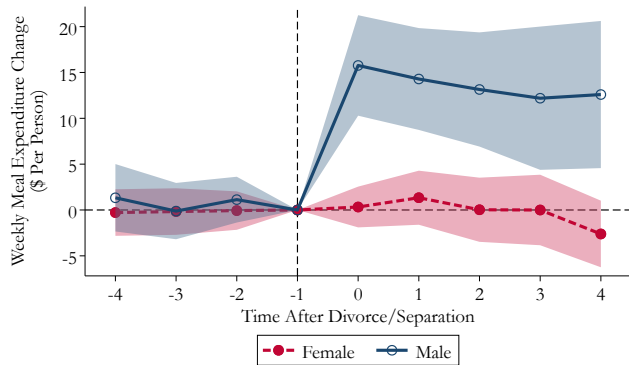
The same events in Australia (HILDA)



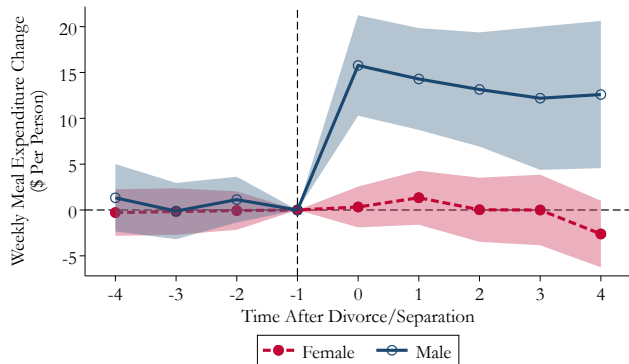
- At cohabitation, only women's housework rises; men's does not change. After divorce, women's falls and men's rises slightly

HILDA waves 1-20, hours per week.

Weekly spending on meals



Weekly spending on meals



- Per-person weekly spending on meals outside the home, around divorce or separation
- Men's spending jumps at the split and stays up; women's is unchanged

HILDA waves 1-20; 463 men, 537 women.

A gendered friction, and what it does to marriage

- Proposed friction: men pay a utility cost γ for housework, **only** when partnered with a woman
 - “performing household chores generates disutility only when there is a woman in the household who can perform them”
 - Makes the allocation of housework unresponsive to wages

A gendered friction, and what it does to marriage

- Proposed friction: men pay a utility cost γ for housework, **only** when partnered with a woman
 - “performing household chores generates disutility only when there is a woman in the household who can perform them”
 - Makes the allocation of housework unresponsive to wages
- **Implication for marriage:** the gains from marriage come from specialization. If the high-earning wife still does the housework, the couple leaves earnings on the table $\rightarrow$ less surplus from marriages where she earns more $\rightarrow$ fewer of them form

A gendered friction, and what it does to marriage

- Proposed friction: men pay a utility cost γ for housework, **only** when partnered with a woman
 - “performing household chores generates disutility only when there is a woman in the household who can perform them”
 - Makes the allocation of housework unresponsive to wages
- **Implication for marriage:** the gains from marriage come from specialization. If the high-earning wife still does the housework, the couple leaves earnings on the table $\rightarrow$ less surplus from marriages where she earns more $\rightarrow$ fewer of them form

Reconciling domestic and market work

1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

Reconciling domestic and market work

1. **Basic facts: female labor force participation across countries and over time**
2. **Location based challenges: seclusion**
Andrew and Smurra (2026), "Seclusion and Women's Work: Descriptive Evidence from India", *Economic Development and Cultural Change*, forthcoming
3. **Children based challenges: child penalties**
Kleven, Landais and Leite-Mariante (2025), "The Child Penalty Atlas", *Review of Economic Studies*
4. **Who does the housework when she earns more?**
Hancock, Lafortune and Low (2025), "Winning the Bread and Baking It Too: Gendered Frictions in the Allocation of Home Production", NBER WP 33393
5. **Discussion: Ho, Jalota and Karandikar**, "Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal" (discussion lead: Lisa)
6. **Experiment: Provision of Childcare**
Caria, Crépon, Krafft and Nagy (2025), "The Barriers to Female Employment: Experimental Evidence from Egypt"

Egypt: mothers say childcare is the barrier

- Egypt, 2023: 15% of women employed vs. 69% of men
- 98% of men and 85% of women say feeding, bathing and changing children should be the mother's responsibility

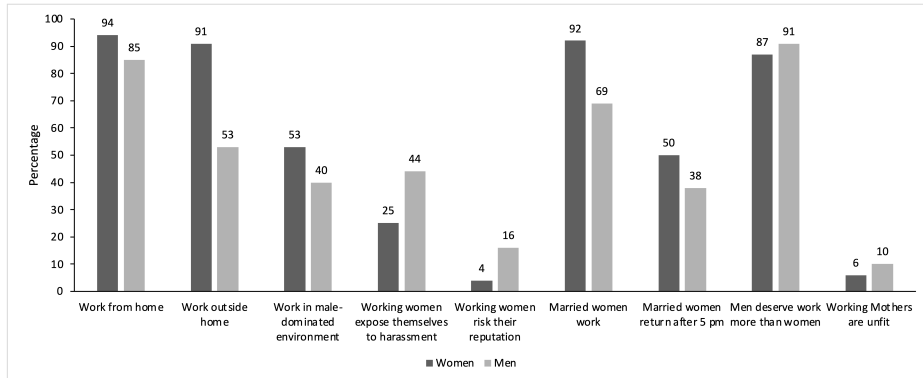
Egypt: mothers say childcare is the barrier

- Egypt, 2023: 15% of women employed vs. 69% of men
- 98% of men and 85% of women say feeding, bathing and changing children should be the mother's responsibility



- Baseline sample (mothers of young children in low-income Greater Cairo): 10.9% employed; 40% of those not employed want to work; 46% of those name childcare as the main barrier

Husbands are much less accepting than wives of women's work



- Work outside the home is acceptable: 91% of women vs. 53% of men. If married: 92% vs. 69%
- Not acceptable to leave a child at a nursery so the mother can work: 35% of women vs. 62% of husbands

Design: a 2 × 2 experiment with 5,047 mothers

- Mothers with a child aged 1–5, low-income areas of Greater Cairo, within 2 km of one of 60 NGO nurseries
- Randomized at the individual level, stratified in blocks of 8
- **Nursery voucher** for one year, for all children 0–5
 - 25% or 75% of the fee; raised to **100%** after take-up stalled at 1–4%
- **Job matching** through private online job platforms: at least 3 vacancies matched to her stated preferences (location, occupation, salary, hours), with help arranging interviews, in 3 rounds
- **Both**, and a control group
- Baseline Dec 2020–Mar 2022 (COVID); two midlines; endline mid-2023

Design: a 2×2 experiment with 5,047 mothers

- Mothers with a child aged 1–5, low-income areas of Greater Cairo, within 2 km of one of 60 NGO nurseries
- Randomized at the individual level, stratified in blocks of 8
- **Nursery voucher** for one year, for all children 0–5
 - 25% or 75% of the fee; raised to **100%** after take-up stalled at 1–4%
- **Job matching** through private online job platforms: at least 3 vacancies matched to her stated preferences (location, occupation, salary, hours), with help arranging interviews, in 3 rounds
- **Both**, and a control group
- Baseline Dec 2020–Mar 2022 (COVID); two midlines; endline mid-2023

$$y_i = \alpha + \beta_1 E_i + \beta_2 V_i + \beta_3 (E_i \times V_i) + X_i' \delta + \text{strata FE} + \varepsilon_i$$

- E_i : job matching; V_i : voucher; β_3 : are the two barriers complements?

Take-up was very low

	(1) Take up nursery voucher	(2) Account creation	(3) Job application submission	(4) Joint take up
	b/se	b/se	b/se	b/se
Employment services	0.000 (0.001)	0.474 (0.014)	0.212 (0.012)	-0.000 (0.000)
100% discount	0.112 (0.009)	-0.001 (0.001)	-0.000 (0.001)	-0.000 (0.000)
100% discount*employment services	-0.010 (0.012)	0.018 (0.020)	0.014 (0.016)	0.026 (0.004)
Strata dummy variables	Yes	Yes	Yes	Yes
Mean of control group	0.000	0.000	0.000	0.000
H1	0.940	0.000	0.000	0.570
H2	0.000	0.636	0.748	0.512
H3	0.425	0.363	0.380	0.000
N	5047	5047	5047	5047

- **Free nursery:** 11% ever used it. Only 3% of non-users cite cost
- **Job matching:** 47% created a profile, 21% applied to at least one job; mothers declined about half of interview invitations
- **4 mothers** were hired through the platforms

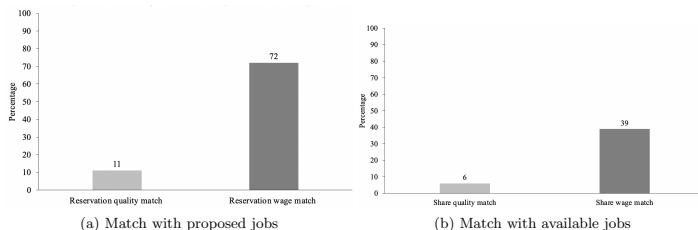
No effect on employment, hours or income

	(1) Work activity	(2) Hours of work (7 days)	(3) Monthly labor income	(4) Employment index
	b/se	b/se	b/se	b/se
Employment services	-0.0103 (0.00970)	-0.541 (0.401)	52.84 (52.85)	-0.00381 (0.0353)
100% discount	-0.00520 (0.0101)	-0.156 (0.422)	-20.68 (28.06)	-0.0189 (0.0292)
100% discount*employment services	0.0113 (0.0140)	0.557 (0.583)	-17.23 (62.99)	0.0204 (0.0465)
Midline 2 dummy variable	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Nursery dummy variables	Yes	Yes	Yes	Yes
Strata dummy variables	Yes	Yes	Yes	Yes
Mean of control group	0.102	3.757	152.896	0.000
H1 (p-value)	0.288	0.178	0.317	0.914
H1 (sharpened q-value)				0.914
H2 (p-value)	0.606	0.711	0.461	0.519
H2 (sharpened q-value)				0.854
H3 (p-value)	0.420	0.339	0.785	0.661
H3 (sharpened q-value)				0.854
N of Observations	8085	8085	8085	8085
N of Individuals	4455	4455	4455	4455

- Job matching: –1.0 pp on a control mean of 10.2% (95% CI about –2.9 to +0.9 pp); voucher: –0.5 pp; no complementarity
- Still null at endline. For comparison, a childcare voucher experiment in Nairobi raised mothers' employment by 17% (**Clark et al. 2019**)

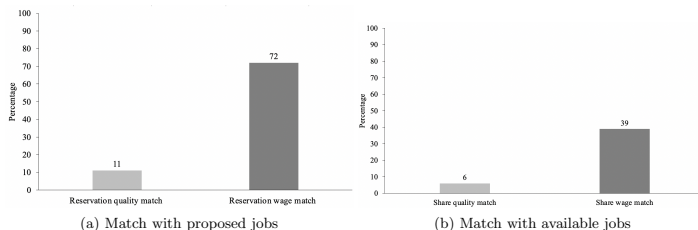
CI is my calculation from the reported s.e.

Why? The jobs on offer are not the jobs mothers would take



- 72% of mothers were offered a job paying at least their reservation wage; only **11%** a job meeting most of their non-wage criteria
- **Part-time:** wanted by 80%; under 1% of vacancies. **Public sector:** the most acceptable job; never offered (all platform jobs are private)

Why? The jobs on offer are not the jobs mothers would take



- 72% of mothers were offered a job paying at least their reservation wage; only **11%** a job meeting most of their non-wage criteria
- **Part-time:** wanted by 80%; under 1% of vacancies. **Public sector:** the most acceptable job; never offered (all platform jobs are private)
- In application decisions, one more matched non-wage attribute is worth about 1.65 log points of salary relative to the reservation wage

And husbands: job matching backfires in non-progressive couples

- “Progressive couple”: both spouses say it is acceptable for women to work outside the home
 - Among those offered job matching, progressive couples are 39% more likely to apply for a job

And husbands: job matching backfires in non-progressive couples

- “Progressive couple”: both spouses say it is acceptable for women to work outside the home
 - Among those offered job matching, progressive couples are 39% more likely to apply for a job
- In **non-progressive** couples, job matching:
 - Raises disagreement between spouses about her working by **17.8 pp** (control mean 28%)
 - Lowers her belief that her husband will become more supportive by 10.6 pp (control mean 43%)
- No such effects in progressive couples

And husbands: job matching backfires in non-progressive couples

- “Progressive couple”: both spouses say it is acceptable for women to work outside the home
 - Among those offered job matching, progressive couples are 39% more likely to apply for a job
- In **non-progressive** couples, job matching:
 - Raises disagreement between spouses about her working by **17.8 pp** (control mean 28%)
 - Lowers her belief that her husband will become more supportive by 10.6 pp (control mean 43%)
- No such effects in progressive couples
- **Main takeaway:** relaxing the barriers mothers name (cost of childcare, finding job openings) does nothing when the binding constraints are the *attributes* of available jobs and husbands' norms

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?
- **Mismatch is measured from stated preferences**, which may already reflect norms and what women expect to be feasible; for the first two rounds, “rejected” jobs are imputed from other mothers’ applications

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?
- **Mismatch is measured from stated preferences**, which may already reflect norms and what women expect to be feasible; for the first two rounds, “rejected” jobs are imputed from other mothers’ applications
- **Norms are not randomized:** “progressive couple” is correlated with education and job prospects; the backfire result is heterogeneity on a non-randomized moderator, in the 1,575 couples where both spouses answered

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?
- **Mismatch is measured from stated preferences**, which may already reflect norms and what women expect to be feasible; for the first two rounds, “rejected” jobs are imputed from other mothers’ applications
- **Norms are not randomized:** “progressive couple” is correlated with education and job prospects; the backfire result is heterogeneity on a non-randomized moderator, in the 1,575 couples where both spouses answered
- **What was actually offered:** private-sector, mostly full-time jobs, plus nurseries that close at 1–3pm. It cannot distinguish “no acceptable jobs exist in Cairo” from “none on these platforms”

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?
- **Mismatch is measured from stated preferences**, which may already reflect norms and what women expect to be feasible; for the first two rounds, “rejected” jobs are imputed from other mothers’ applications
- **Norms are not randomized:** “progressive couple” is correlated with education and job prospects; the backfire result is heterogeneity on a non-randomized moderator, in the 1,575 couples where both spouses answered
- **What was actually offered:** private-sector, mostly full-time jobs, plus nurseries that close at 1–3pm. It cannot distinguish “no acceptable jobs exist in Cairo” from “none on these platforms”
- **Design changes and attrition:** subsidy levels changed mid-study; 36% endline attrition

Caria et al.: what to push on

- **A null from non-take-up:** 4 hires and 11% nursery use leave little power to learn what jobs or nurseries do for users. Is this a paper about why demand for these services is low?
- **Mismatch is measured from stated preferences**, which may already reflect norms and what women expect to be feasible; for the first two rounds, “rejected” jobs are imputed from other mothers’ applications
- **Norms are not randomized:** “progressive couple” is correlated with education and job prospects; the backfire result is heterogeneity on a non-randomized moderator, in the 1,575 couples where both spouses answered
- **What was actually offered:** private-sector, mostly full-time jobs, plus nurseries that close at 1–3pm. It cannot distinguish “no acceptable jobs exist in Cairo” from “none on these platforms”
- **Design changes and attrition:** subsidy levels changed mid-study; 36% endline attrition
- **Next step:** vary the job attributes themselves (part-time, near home, from home) — as in our discussion paper